

WATERTON PARK GATE, AB, Canada

WMO: 711540

Lat: 49.1311N

Lon: 113.8086W

Elev: 4229

StdP: 12.58

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.0	(c) -12.5	(d) -26.3	(e) 1.5	(f) -19.7	(g) -18.2	(h) 2.4	(i) -11.9	(j) 44.8	(k) 40.3	(l) 41.8	(m) 35.5	(n) 3.2	(o) 40	(p) 0.767

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 27.4	(c) 84.9	(d) 59.1	(e) 81.7	(f) 58.2	(g) 78.4	(h) 57.4	(i) 62.7	(j) 77.5	(k) 60.8	(l) 76.8	(m) 59.0	(n) 74.8	(o) 11.0	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.2	81.6	69.5	54.7	74.6	66.4	52.5	68.7	64.1	30.5	77.3	29.0	77.0	27.6	74.9	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 36.4	(b) 32.0	(c) 28.4	(e) -31.5	(f) 90.1	(g) 9.4	(h) 3.5	(i) -38.3	(j) 92.6	(k) -43.8	(l) 94.7	(m) -49.1	(n) 96.7	(o) -56.0	(p) 99.3		
(4) 36.4	(5) 32.0	(6) 28.4	(7) -31.8	(8) 66.6	(9) 9.3	(10) 1.8	(11) -38.5	(12) 67.9	(13) -43.9	(14) 69.0	(15) -49.2	(16) 70.0	(17) -55.9	(18) 71.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	24.4	24.6	30.0	39.2	48.2	54.8	61.9	60.4	53.0	42.5	32.0	24.7
	DBStd	17.67	15.90	14.93	14.29	9.63	8.36	6.39	5.57	6.13	8.60	10.46	13.87	14.50
	HDD50	4292	793	712	620	338	135	24	2	5	64	270	546	783
	HDD65	8673	1257	1132	1083	774	523	310	128	164	366	698	990	1248
	CDD50	1158	1	0	1	14	78	169	369	328	155	37	6	0
	CDD65	65	0	0	0	0	1	5	30	23	6	0	0	0
	CDH74	1825	0	0	0	1	35	115	778	699	192	5	0	0
	CDH80	458	0	0	0	0	2	23	221	180	31	1	0	0
Wind	WSAvg	12.2	16.7	14.2	12.8	11.3	10.3	10.0	8.4	7.9	9.6	13.4	15.6	16.8
Precipitation	PrecAvg	33.4	2.7	2.2	2.6	3.1	3.8	5.0	2.2	2.3	2.6	2.8	3.0	2.4
	PrecMax	46.3	6.9	4.1	3.8	5.6	9.1	11.5	7.5	5.8	5.8	6.7	5.4	4.1
	PrecMin	25.5	1.2	0.7	0.9	1.5	0.8	1.0	0.0	0.6	0.3	1.1	1.5	1.5
	PrecStd	5.2	1.2	0.7	0.7	0.9	1.9	2.8	1.6	1.4	1.3	1.2	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	53.5	58.9	69.3	78.3	84.0	89.8	88.9	83.9	72.3	61.2	51.6
		MCWB	40.8	39.3	43.0	48.7	54.2	60.7	60.6	58.6	56.9	50.7	44.3	40.6
	2%	DB	48.1	48.1	53.6	63.9	73.3	76.9	85.5	84.6	79.3	65.9	55.5	46.8
		MCWB	37.4	37.2	41.2	45.8	52.4	57.4	59.9	58.1	55.5	49.1	42.5	37.0
	5%	DB	44.1	44.4	49.9	58.7	68.4	73.0	82.1	81.4	75.3	60.9	51.3	43.2
		MCWB	35.9	35.3	39.0	43.5	50.4	55.7	59.2	57.6	54.0	46.4	40.7	35.4
	10%	DB	41.5	41.3	46.4	53.7	63.8	69.0	78.3	77.9	70.2	56.9	47.6	40.2
		MCWB	34.5	33.8	37.2	41.1	48.3	53.4	58.9	57.2	52.1	44.5	38.7	33.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	40.7	44.7	49.8	56.1	63.5	66.0	63.8	60.3	53.2	47.3	41.7
		MCDB	51.2	49.9	55.9	67.3	74.1	77.1	79.8	78.6	76.6	67.9	57.3	49.6
	2%	WB	38.8	38.5	42.0	46.6	53.6	59.6	63.5	61.5	57.3	49.9	44.2	38.7
		MCDB	45.5	46.1	51.2	61.8	69.7	74.1	78.1	77.6	74.7	64.2	54.4	44.4
	5%	WB	36.6	36.2	39.9	44.1	51.4	57.0	61.7	59.7	55.1	47.5	41.3	36.2
		MCDB	42.9	43.7	48.8	56.8	66.6	70.6	77.1	76.3	72.1	59.5	50.4	42.0
	10%	WB	34.9	34.0	37.7	41.9	49.3	54.6	59.9	58.0	53.0	45.1	39.1	34.0
		MCDB	41.1	41.0	45.8	52.8	62.2	66.2	75.6	74.4	68.3	55.8	46.4	40.0
Mean Daily Temperature Range	5% DB	MDBR	16.6	17.0	17.9	18.5	20.1	20.9	27.4	28.4	23.7	17.6	14.6	14.1
		MCDBR	15.0	16.0	19.3	26.4	28.5	29.6	35.7	37.4	34.6	23.3	16.3	13.3
		MCWBR	9.6	10.0	11.0	13.7	13.5	14.8	15.7	16.1	15.8	12.2	9.5	8.9
	5% WB	MCDBR	14.0	16.1	18.8	24.6	25.7	27.7	31.6	32.6	32.4	23.6	15.9	13.8
		MCWBR	9.9	11.0	11.7	13.4	12.8	15.0	16.3	15.8	15.7	13.0	10.0	10.1
Clear-Sky Solar Irradiance	taub	0.238	0.250	0.276	0.337	0.342	0.334	0.331	0.346	0.315	0.281	0.255	0.229	
	taud	2.384	2.367	2.365	2.252	2.374	2.449	2.470	2.420	2.486	2.561	2.498	2.401	
	Ebn at Noon	270	292	300	288	291	293	292	283	283	278	262	259	
	Edh at Noon	20	26	31	39	36	34	33	33	28	22	18	17	
All-Sky Solar Radiation	RadAvg	377	689	1061	1483	1773	1962	2230	1801	1271	774	415	299	
	RadStd	33	55	98	106	152	159	141	120	114	78	34	28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page